

CS & IT ENGINEERING



Operating System

Disk Scheduling Algorithm

One Shot



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Recap of Previous Lecture



doubt?

Topic

Blocks and File Allocation Methods

Topic

Unix i-node



Topics to be Covered



Topic

Disk Scheduling Algorithm

Topic

SSTF, Scan, Look



Topic : Question

[GATE-2022]



#Q. Consider two files systems A and B , that use contiguous allocation and linked allocation, respectively. A file of size 100 blocks is already stored in A and also in B. Now, consider inserting a new block in the middle of the file (between 50th and 51st block), whose data is already available in the memory. Assume that there are enough free blocks at the end of the file and that the file control blocks are already in memory. Let the number of disk accesses required to insert a block in the middle of the file in A and B are n_A and n_B respectively, then the value of $n_A + n_B$ is 153?

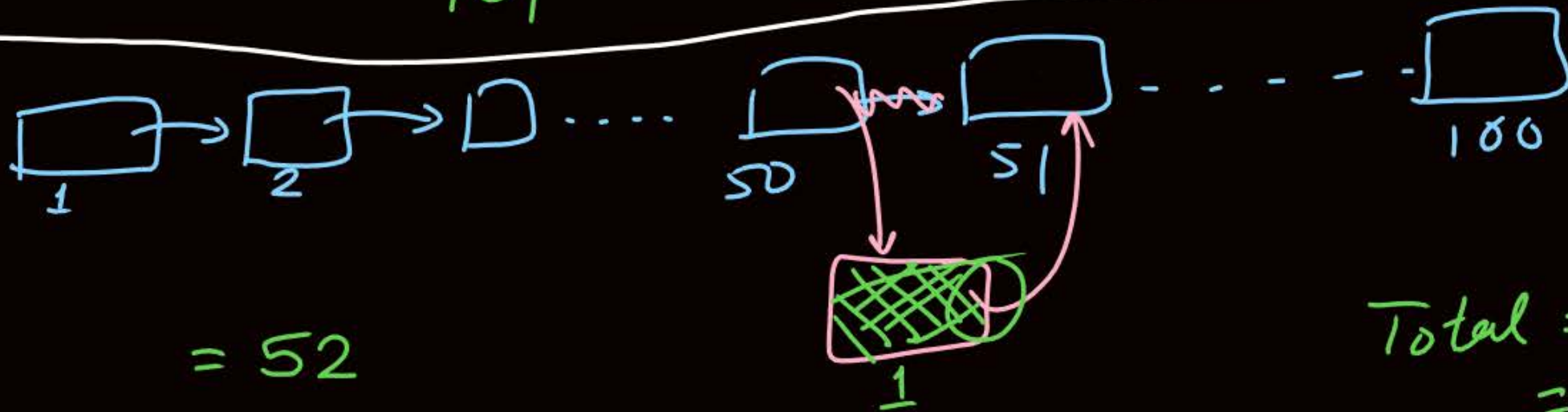


Contiguous :-



block accesses = 100 (for shifting 50 blocks 1 step ahead)
1 (to insert content)
101

Linked :-



= 52

Total = 101 + 52
= 153



#Q. In a computer system, four files of size 11050 bytes, 4990 bytes, 5170 bytes and 12640 bytes need to be stored. For storing these files on disk, we can use either 100 byte disk blocks or 200 byte disk blocks (but can't mix block sizes). For each block used to store a file, 4 bytes of bookkeeping information also needs to be stored on the disk. Thus, the total space used to store a file is the sum of the space taken to store the file and the space taken to store the book keeping information for the blocks allocated for storing the file. A disk block can store either bookkeeping information for a file or data from a file, but not both.

What is the total space required for storing the files using 100 byte disk blocks and 200 byte disk blocks respectively?



35400 and 35800 bytes



35800 and 35400 bytes



35600 and 35400 bytes



35400 and 35600 bytes



Topic : Solution

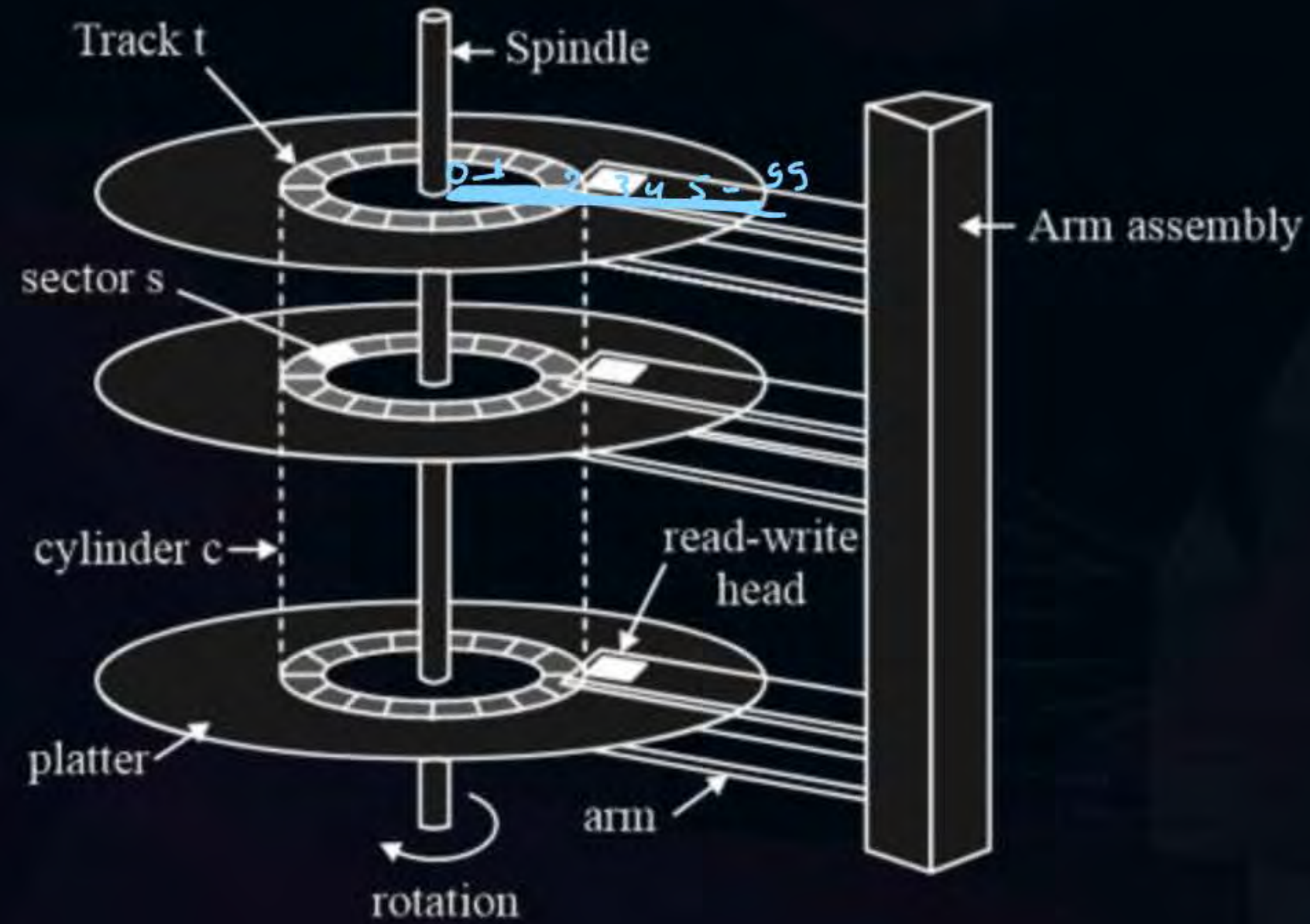
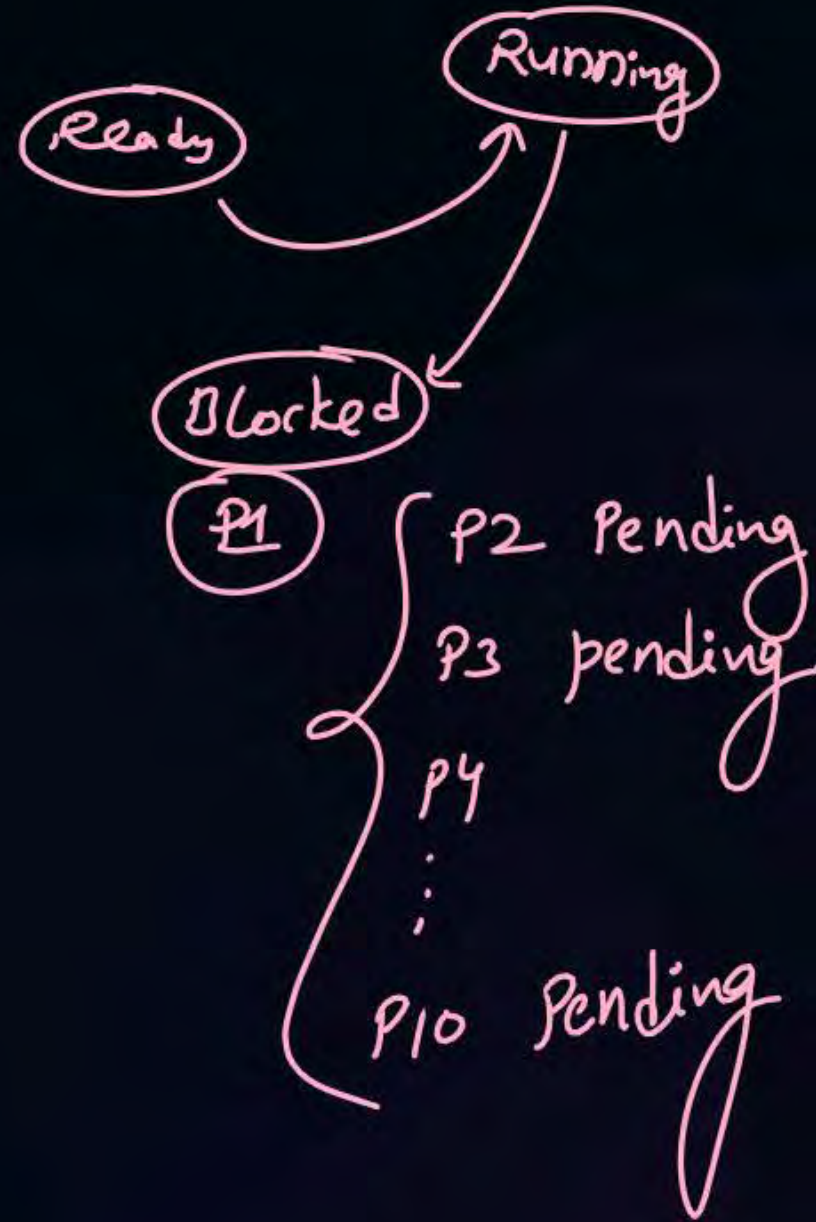
File Size	No. of blocks to store file	No. of blocks to store bookkeeping	Total blocks	Total Size
11050 Bytes	$\left\lceil \frac{11050\text{B}}{100\text{B}} \right\rceil = 111$	$\left\lceil \frac{111 * 4\text{B}}{100\text{B}} \right\rceil = 5$	116	
4990 Bytes	$\left\lceil \frac{4990}{100} \right\rceil = 50$	$\left\lceil \frac{50 * 4\text{B}}{100\text{B}} \right\rceil = 2$	52	
5170 Bytes	$\left\lceil \frac{5170}{100} \right\rceil = 52$	$\left\lceil \frac{52 * 4\text{B}}{100\text{B}} \right\rceil = 3$	55	
12640 Bytes	$\left\lceil \frac{12640}{100} \right\rceil = 127$	$\left\lceil \frac{127 * 4\text{B}}{100} \right\rceil = 6$	133	

Total 356

Total size = $356 * 100\text{B}$
 $= 35600\text{B}$



Topic : Disk





Topic : Cylinder



- Collection of tracks of same radius from all surfaces





Topic : Disk Scheduling



- Done by operating systems to schedule I/O requests arriving for the disk

↓
performance of disk improves
↓





Topic : Disk Scheduling Algorithms

1. FCFS (First Come First Serve)
2. SSTF (Shortest Seek Time First)
3. Scan
4. C-Scan (Circular-Scan)
5. Look
6. C-Look (Circular-Look)



Topic : FCFS (First Come First Serve)

Suppose total 100 cylinders 0-99

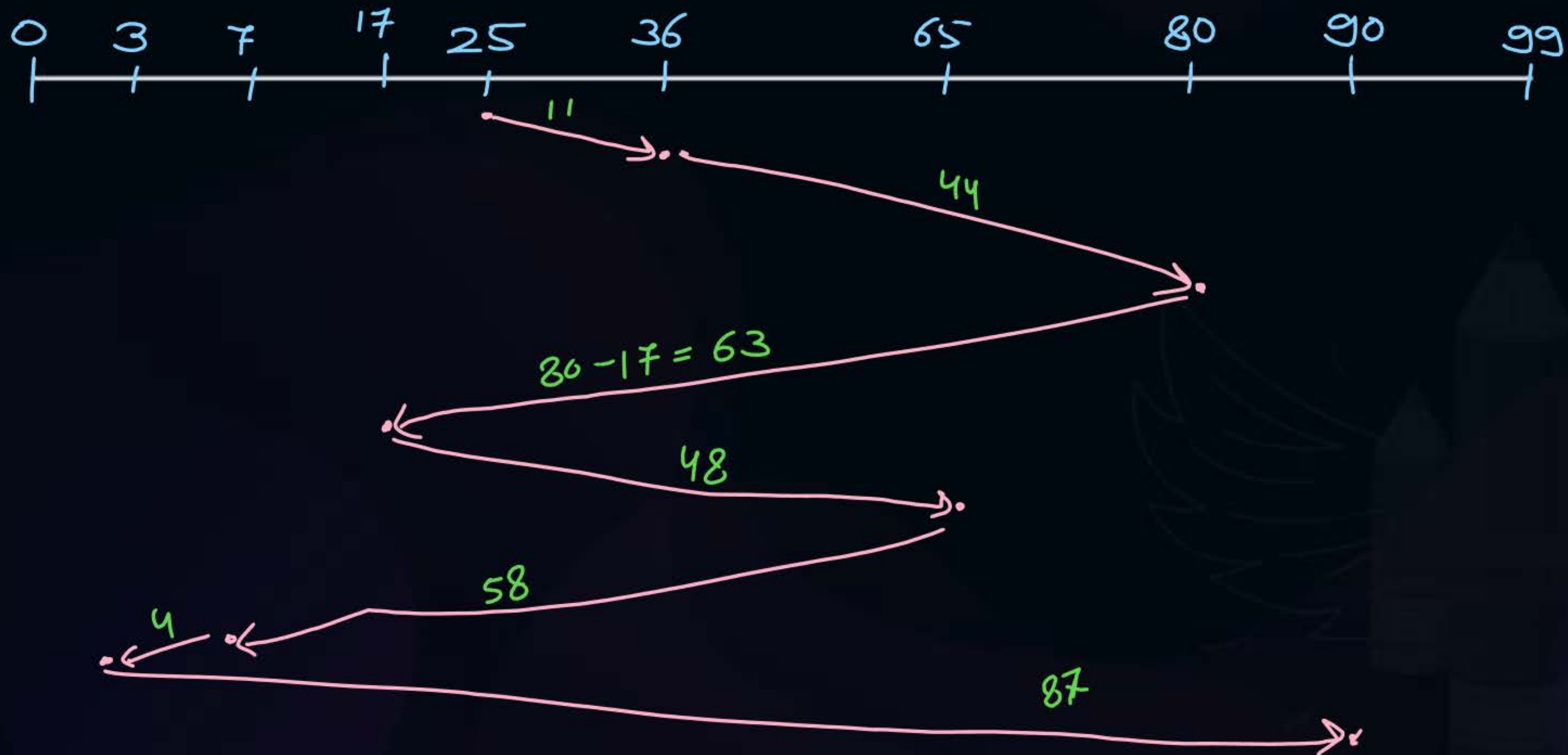
The order of request is: 36, 80, 17, 65, 7, 3, 90

The Read/Write arm is at 25



Topic : FCFS (First Come First Serve)

Suppose the order of request is: 36, 80, 17, 65, 7, 3, 90



Total no. of head movements = 315

no. of times R/W head changed its direction = 4

Ques) If per head movement, disk takes = 2 ms seek time
Total seek time for servicing all requests = 630 ms?

Solⁿ

$$= 315 * 2 \text{ ms}$$

$$= 630 \text{ ms}$$

ans) 1 head movement = 2ms
each direction change takes = 10ms
total seek time = 670 ms

solⁿ

$$= (315 * 2\text{ms}) + (4 * 10\text{ms})$$
$$= 670\text{ms}$$



Topic : FCFS (First Come First Serve)

Advantages:

- Every request gets a fair chance
- No indefinite postponement (*starvation*)

Disadvantages:

- Does not try to optimize seek time
- May not provide the best possible service



Topic : SSTF (Shortest Seek Time First)

→ service the nearest request first.



Suppose total 100 cylinders 0-99

The order of request is: 36, 80, 17, 65, 7, 3, 90

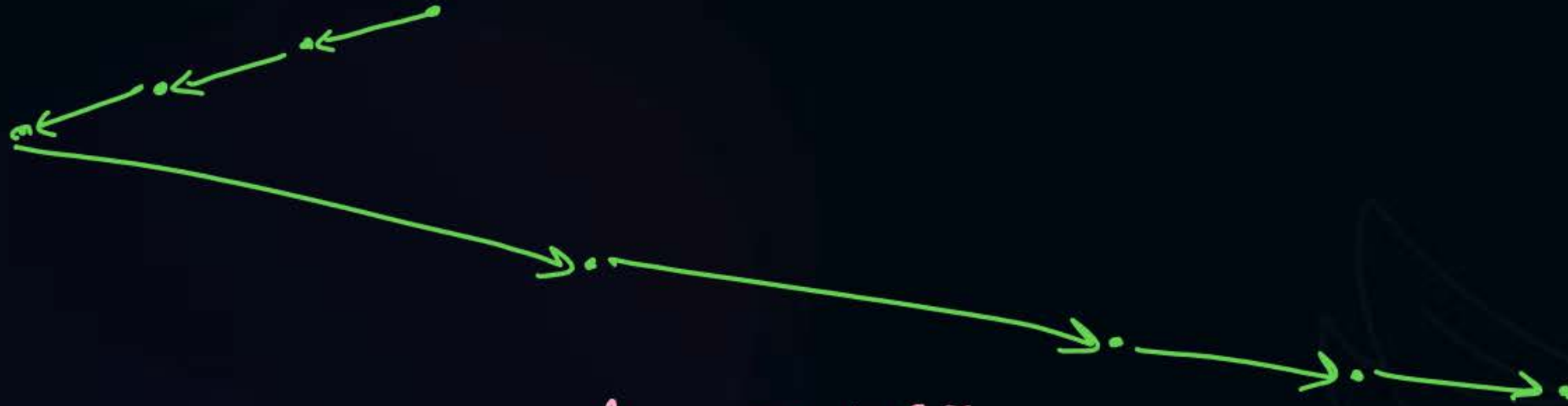
The Read/Write arm is at 25





Topic : SSTF (Shortest Seek Time First)

Suppose the order of request is: 36, 80, 17, 65, 7, 3, 90



$$\begin{aligned}\text{no. of head movements} &= 22 + 87 \\ &= 109\end{aligned}$$



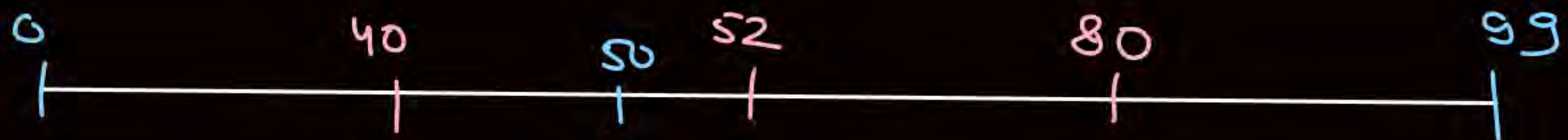
Topic : SSTF (Shortest Seek Time First)

Advantages:

- Average Response Time decreases
- Throughput increases

Disadvantages:

- Overhead to calculate seek time in advance
- Can cause Starvation for a request if it has higher seek time as compared to incoming requests
- High variance of response time as SSTF favors only some requests





Topic : Scan (Elevator)

Suppose total 100 cylinders 0-99

The order of request is: 36, 80, 17, 65, 7, 3, 90

The Read/Write arm is at 25

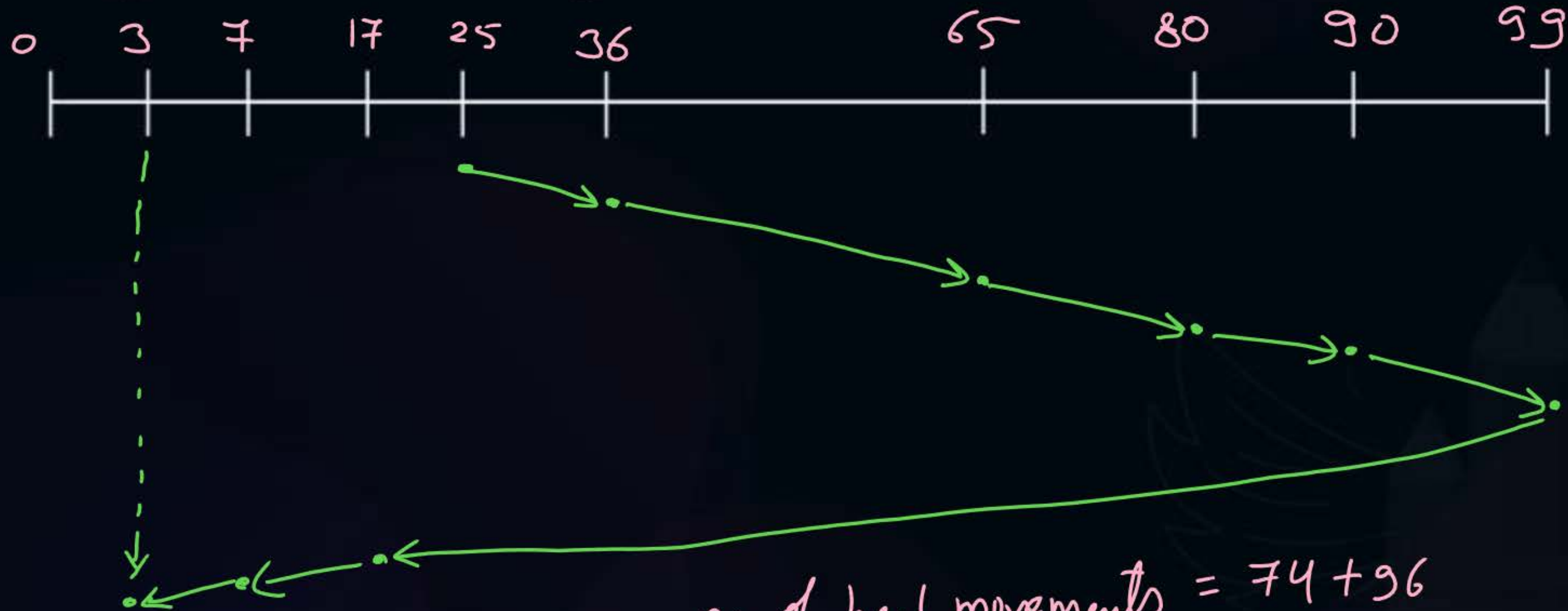
The arm should move “towards the larger value”





Topic : Scan (Elevator)

Suppose the order of request is: 36, 80, 17, 65, 7, 3, 90



$$\begin{aligned}\text{no. of head movements} &= 74 + 96 \\ &= 170\end{aligned}$$



Topic : Scan

Advantages:

- High throughput
- Low variance of response time
- Average response time

Disadvantages:

- Long waiting time for requests for locations just visited by disk arm



Topic : C-Scan

(Circular-Scan)



Suppose total 100 cylinders 0-99

The order of request is: 36, 80, 17, 65, 7, 3, 90

The Read/Write arm is at 25

The arm should move “towards the larger value”

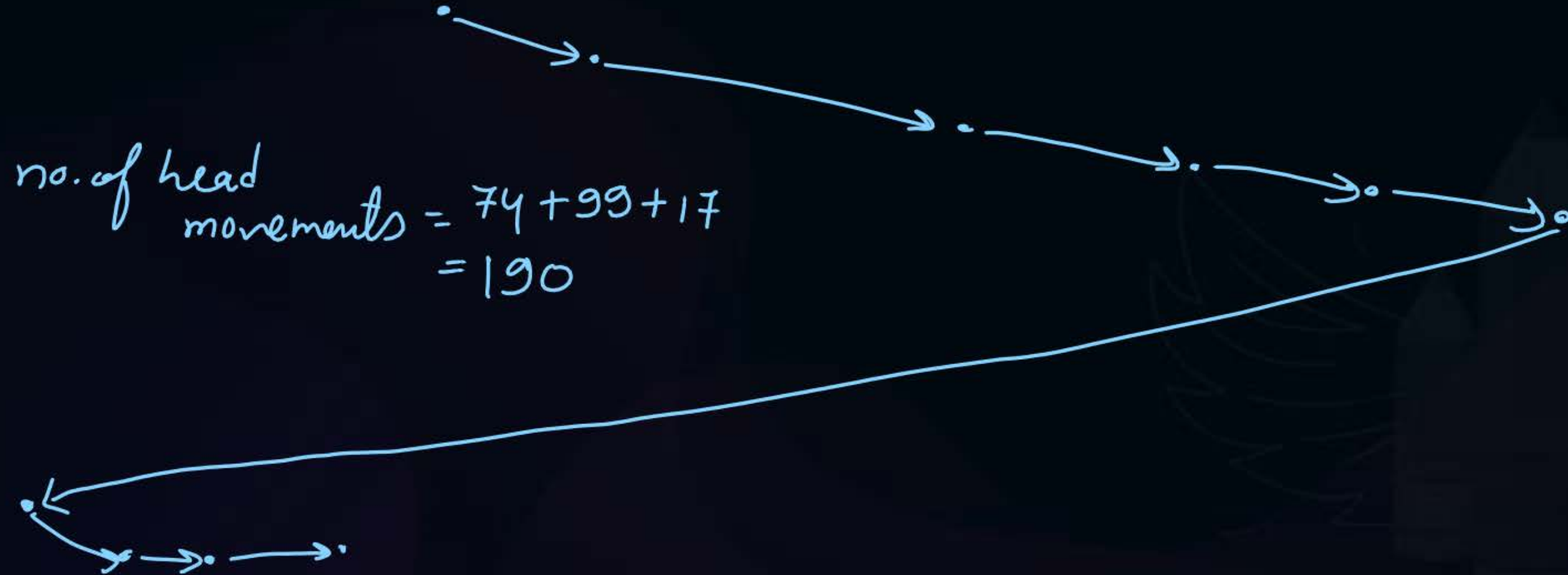




Topic : C-Scan



Suppose the order of request is: 36, 80, 17, 65, 7, 3, 90





Topic : C-Scan



Advantages:

- Provides more uniform wait time compared to SCAN





Topic : Look

Suppose total 100 cylinders 0-99

The order of request is: 36, 80, 17, 65, 7, 3, 90

The Read/Write arm is at 25

The arm should move “towards the larger value”

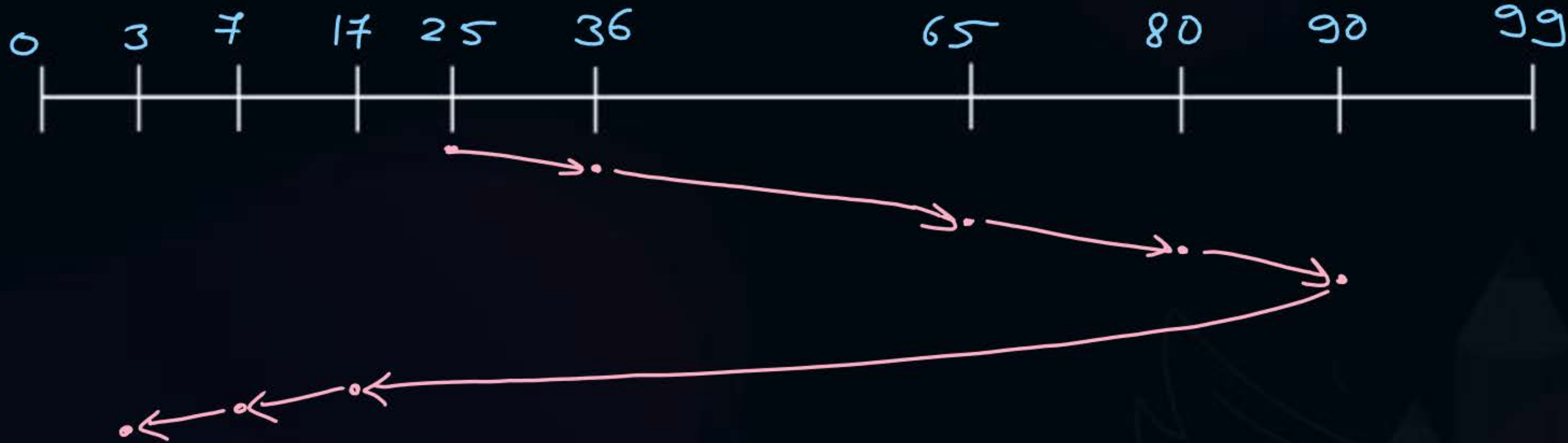




Topic : Look



Suppose the order of request is: 36, 80, 17, 65, 7, 3, 90



$$\begin{aligned}\text{no. of head movement} &= 65 + 87 \\ &= 152\end{aligned}$$



Topic : C-Look

Suppose total 100 cylinders 0-99

The order of request is: 36, 80, 17, 65, 7, 3, 90

The Read/Write arm is at 25

The arm should move “towards the larger value”

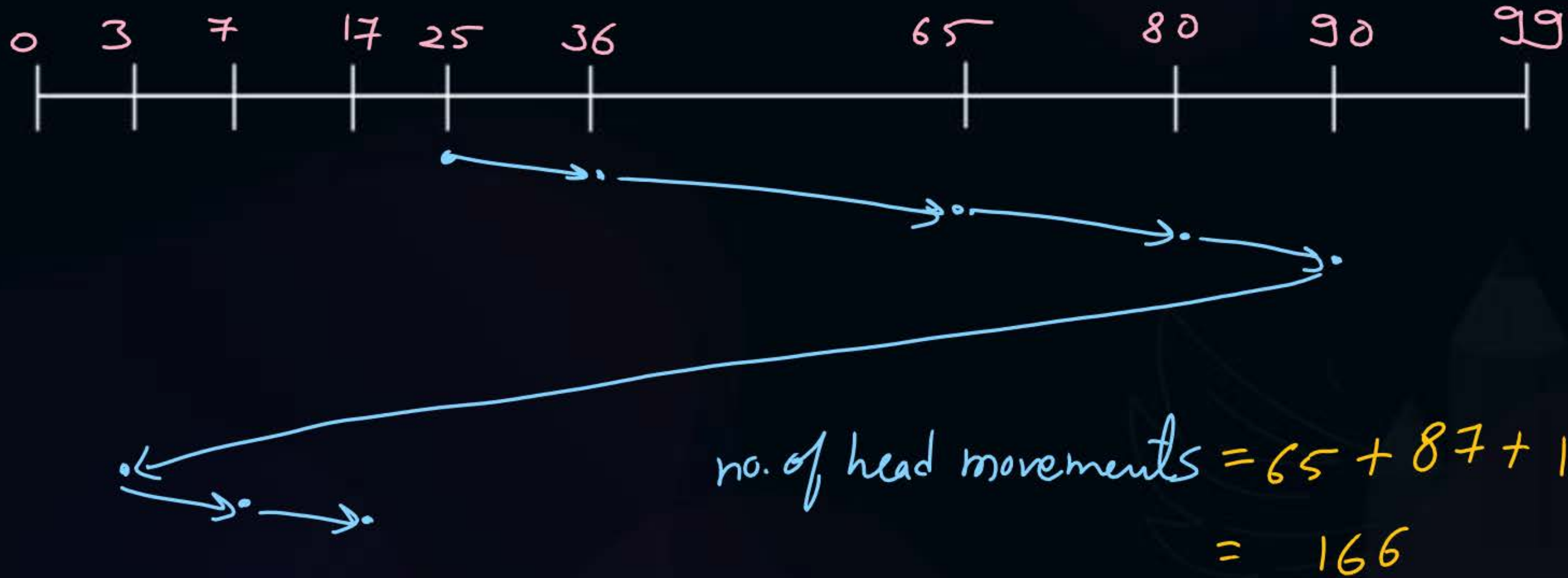




Topic : C-Look



Suppose the order of request is: 36, 80, 17, 65, 7, 3, 90



[MCQ]



#Q. Consider an operating system capable of loading and executing a single sequential user process at a time. The disk head scheduling algorithm used is First Come First Served (FCFS). If FCFS is replaced by Shortest Seek Time First (SSTF), claimed by the vendor to give 50% better benchmark results, what is the expected improvement in the I/O performance of user programs? **[2004]**

unprogramming

A 50%

B 40%

C 25%

D ✓ 0%

#Q. The head of a hard disk serves request following the shortest seek time first (SSTF) policy. The head is initially positioned at track number 180.

Which of the request sets will cause the head to change its direction after servicing every request assuming that the head does not change direction if there is a tie in SSTF and all the request arrive before the servicing starts?

- A** 11, 139, 170, 178, 181, 184, 201, 265
- B** ✓ 10, 138, 170, 178, 181, 185, 201, 265
- C** 10, 139, 169, 178, 181, 184, 201, 265
- D** 10, 138, 170, 178, 181, 185, 200, 265

[2007]



2 mins Summary

Topic

Disk Scheduling Algorithm

Topic

SSTF, Scan, Look





Happy Learning

THANK - YOU